

Functional Requirements Document

Client Onboarding & KYC Workflow Optimization

Document status: Document status: FINAL — FR-003 and FR-004 validated against Day 4 mock-dataset analysis (n=2000). FR-014 added to resolve override precedence. No other sections are affected by that patch.

FR-001 — Risk Score Inputs

The system shall compute a risk score for every applicant using exactly three inputs, each normalized to a 0-100 scale before weighting:

- country_risk_score (0-100, where 100 = highest-risk jurisdiction, per a maintained country risk reference list)
- transaction_velocity_score (0-100, where 100 = highest anomalous velocity, derived from declared transaction frequency/volume at signup)
- verification_match_rate (0-100, where 100 = perfect document/selfie match; lower values indicate lower confidence in identity match)

FR-002 — Score Direction Convention

For all three inputs in FR-001, a *higher* normalized value shall represent *higher* risk. verification_match_rate shall therefore be inverted before weighting (i.e., the system uses 100 - verification_match_rate as the risk-contributing value), since a high match rate indicates *low* risk.

FR-003 — Weighted Score Formula

The system shall calculate risk_score as:

$$\begin{aligned} \text{risk_score} = & (\text{country_risk_score} \times 0.40) \\ & + (\text{transaction_velocity_score} \times 0.30) \\ & + ((100 - \text{verification_match_rate}) \times 0.30) \end{aligned}$$

Validated against mock dataset (Day 4, n=2000): this formula's weights were tested as the working hypothesis for the threshold-tuning exercise (FR-004). The formula correctly produces a continuous, well-distributed risk_score (range 4.4-91.7, mean 34.7) suitable for tiering. However, the formula's separation of true risk from legitimate applicants is moderate, not strong — at no tested threshold does precision exceed ~0.83 without recall falling below 0.03. This is documented as a known model limitation in the case study rather than treated as a solved

problem; a future iteration could test alternative weightings or additional features to improve separation.

FR-004 — Risk Tier Thresholds

The system shall assign a risk_tier based on risk_score using the following thresholds:

- $\text{risk_score} < 50 \rightarrow \text{Low}$
- $50 \leq \text{risk_score} < 75 \rightarrow \text{Medium}$
- $\text{risk_score} \geq 75 \rightarrow \text{High}$

Threshold validated against mock dataset (Day 4, n=2000). A sweep from 20-75 was run, comparing precision/recall/F1 at each candidate (see 07_Scoring_Script/threshold_analysis.py output). Findings:

- F1 peaks at threshold 40 (precision 0.346, recall 0.548), but this would route 32.2% of all applicants to High-tier/EDD — operationally unworkable, since it defeats the business goal (BRD §3) of EDD being applied to a risk-concentrated minority, not a third of all users.
- **Threshold 50 was selected instead**, which routes 15.2% of applicants to High-tier, catching 32.4% of real flagged-risk cases at precision 0.436. This is a deliberate precision-and-volume-leaning choice over the F1-optimal point, prioritizing a workable EDD caseload over maximum recall.
- This is a known tradeoff, not a fully solved one: at threshold 50, the system still misses 67.6% of cases later flagged as risky. This limitation is carried forward explicitly into the case study rather than hidden.
- The original FRD draft's candidate thresholds (70, 75) are **not used** — both were shown to sit far out in the score distribution's tail ($\leq 1.6\%$ of applicants), producing recall too low (≤ 0.057) to be defensible as a risk-tiering control.

FR-005 — PEP/Sanctions Override Rule

If an applicant matches a Politically Exposed Person (PEP) list or a sanctions list at any point during intake, the system shall set risk tier = "High" and route the applicant to Enhanced Due Diligence (EDD), **overriding the result of FR-003/FR-004 regardless of computed risk score**. This override is unconditional: a PEP match with an otherwise low computed score is still routed to EDD.

FR-006 — Low-Tier Routing

If risk tier = "Low" and no PEP/sanctions override (FR-005) applies, the system shall route the

applicant directly to account activation without manual Compliance review, recording the automated decision and the contributing risk score for audit purposes.

FR-007 — Medium-Tier Routing

If risk tier = "Medium" and no PEP/sanctions override applies, the system shall route the applicant to a single standard Compliance review step (one reviewer, standard checklist), not Enhanced Due Diligence.

FR-008 — High-Tier Routing

If risk tier = "High" (by score or by PEP/sanctions override), the system shall route the applicant to Enhanced Due Diligence: minimum two-reviewer sign-off, full source-of-funds documentation request, and mandatory sanctions re-screen at decision time (not just at intake).

FR-009 — Verification Match Rate Failure Handling

If `verification_match_rate < 40` (regardless of computed risk score or tier), the system shall force a re-upload request before proceeding to tier routing, capped at 2 retry attempts. After 2 failed attempts, the applicant shall be routed to manual Compliance review regardless of tier.

FR-010 — Expired ID Handling

If the uploaded ID document's expiry date is in the past at time of upload, the system shall reject the document immediately (before any verification match calculation) and prompt the applicant for a valid, unexpired document. This check shall occur prior to FR-001's input calculation, not after.

FR-011 — Audit Trail Requirement

For every applicant, the system shall persist: all three FR-001 input values, the computed risk score, the assigned risk tier, whether PEP/sanctions override applied, the routing decision, and a timestamp — regardless of final outcome. This record must be sufficient to reconstruct, after the fact, exactly why a given applicant received a given tier.

FR-012 — Threshold Change Control

Any future change to the FR-004 threshold values shall require: (a) a documented analysis showing the false-positive/false-negative impact of the proposed change on a representative dataset, and (b) Compliance sign-off, since this value directly governs EDD coverage. This requirement formalizes that thresholds are not to be changed informally post-launch.

FR-013 — Error Handling: Verification Service Unavailable

If the document/selfie verification step (referenced in the As-Is/To-Be diagrams) fails to return a result due to a downstream error, the system shall not default the applicant to "Low" risk by omission. The applicant shall be held in a pending verification state and routed to manual review after a defined timeout (exact timeout value: TBD, not in scope for this portfolio project's mock dataset, but the *rule* — never default to Low on missing data — is a hard requirement).

FR-014 — Override Precedence Order

When multiple rules in this document apply to the same applicant simultaneously, the system shall resolve them in this fixed order:

1. **FR-010 (expired ID) and FR-009 (verification match rate < 40)** — input-quality gates. These fire before any routing decision is made, since routing cannot be meaningfully evaluated against unreliable or invalid input. An applicant who fails these checks is held at re-upload/retry per FR-009/FR-010 regardless of any other condition that may also apply to them.
2. **FR-005 (PEP/sanctions override)** — fires once input quality is resolved (or retry limits are exhausted per FR-009). A PEP/sanctions match forces `risk_tier = "High"` and EDD routing, overriding whatever FR-003/FR-004 would otherwise compute.
3. **FR-003/FR-004 (weighted score and tier thresholds)** — fires only if no override from steps 1-2 applies.

Worked example: An applicant has a PEP match and a verification match rate of 32 (below the FR-009 threshold of 40). The system does not skip straight to EDD. It first applies FR-009: the applicant is prompted to re-upload (up to 2 attempts). Only after the verification step is resolved — either a passing match rate is obtained, or the 2-attempt limit is exhausted and the applicant is escalated per FR-009's own fallback — does the system apply FR-005 and confirm the destination is EDD (which, given the PEP match, it always will be, regardless of what the eventual match rate turns out to be). This ordering exists because EDD review depends on having the *best available* identity evidence, arguably more than a standard review does — a PEP-flagged applicant with unverified identity documents is not a case where Compliance should accept degraded input just because the destination was already decided.